

## Test Case 1: Donor Creation(Unit Testing )

| Test Case ID     | TC_Donor_001                                                                                   |
|------------------|------------------------------------------------------------------------------------------------|
| Module           | Blood Donor Management                                                                         |
| Test Case Title  | Verify successful creation of a blood donor                                                    |
| Objective        | To ensure the <b>BloodDonor</b> model stores valid data correctly                              |
| Precondition     | Django project and test environment are set up                                                 |
| Test Steps       | 1. Create a donor object with valid details2. Retrieve donor from DB3. Assert data correctness |
| Input Data       | name: John Doe, age: 30, gender: Male, blood group: O+, contact: 9876543210                    |
| Expected Result  | Donor is saved and blood group is correctly stored                                             |
| Actual Result    | Donor was saved and values matched                                                             |
| Status           | Pass                                                                                           |
| Test Environment | Python 3.10, Django 4.x, PostgreSQL, PyCharm                                                   |

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Ran 1 test in 0.200s

OK
Destroying test database for alias 'default'...
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Fig 3. Unit Test Output for Donor Creation

## Test Case 2: Blood Request Creation(Integration Testing)

| Test Case ID | TC_Request_001                                             |
|--------------|------------------------------------------------------------|
| Module       | Blood Request                                              |
| Title        | Verify blood request creation                              |
| Objective    | To check if a valid blood request can be created and saved |
| Precondition | Patient or hospital is registered in the system            |
| Test Steps   | 1. Fill the blood request form2. Submit the form           |
| Input Data   | blood group: A+, units: 2, hospital_id: 5                  |

|                        |                                              |
|------------------------|----------------------------------------------|
| <b>Expected Result</b> | Blood request is saved with status "Pending" |
| <b>Actual Result</b>   | As expected                                  |
| <b>Status</b>          | Pass                                         |

### Test Case 3: Hospital Creation(Unit Testing)

|                         |                                                                                                                |
|-------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Test Case ID</b>     | TC_Hospital_006                                                                                                |
| <b>Module</b>           | Hospital Management                                                                                            |
| <b>Test Case Title</b>  | Verify creation of a hospital record                                                                           |
| <b>Objective</b>        | Ensure that hospital details are saved correctly in the database                                               |
| <b>Precondition</b>     | Django environment is set up and migrations applied                                                            |
| <b>Test Steps</b>       | 1. Create a Hospital object with valid details2. Retrieve hospital from DB3.<br>Check field values             |
| <b>Input Data</b>       | Name: City HospitalType: GeneralPhone: 1234567890Address: 123 Main St,<br>DelhiEmail: contact@cityhospital.com |
| <b>Expected Result</b>  | Hospital record is saved with correct field values                                                             |
| <b>Actual Result</b>    | Hospital saved correctly (Pass)                                                                                |
| <b>Status</b>           | Pass                                                                                                           |
| <b>Test Environment</b> | Python 3.10, Django 4.x, PostgreSQL                                                                            |

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Terminal  Local (2) x  Local x  +  v
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Ran 1 test in 0.005s

OK
Destroying test database for alias 'default'...
(.venv) (base) divya@divya-HP-Pavilion-Laptop-14-ec1xxx:
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### Test Case 4: Camp Registration(Integration Testing)

|                     |                                                     |
|---------------------|-----------------------------------------------------|
| <b>Test Case ID</b> | TC_CampReg_001                                      |
| <b>Module</b>       | Camp Registration                                   |
| <b>Title</b>        | Verify donor registration for a camp                |
| <b>Objective</b>    | To ensure a donor can register for an existing camp |
| <b>Precondition</b> | Donor and camp exist                                |

|                        |                                                        |
|------------------------|--------------------------------------------------------|
| <b>Test Steps</b>      | 1. Select camp2. Register donor ID                     |
| <b>Input Data</b>      | donor_id: 12, camp_id: 4                               |
| <b>Expected Result</b> | Registration record created in camp_registration table |
| <b>Actual Result</b>   | As expected                                            |
| <b>Status</b>          | Pass                                                   |

```

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Ran 1 test in 0.008s

OK
Destroying test database for alias 'default'...
(.venv) (base) divya@divya-HP-Pavilion-Laptop-14-ec1xxx:

```

## Test Case 5: User Login Authentication(Authentication Testing)

|                        |                                                       |
|------------------------|-------------------------------------------------------|
| <b>Test Case ID</b>    | <b>TC_Login_001</b>                                   |
| <b>Module</b>          | User Authentication                                   |
| <b>Title</b>           | Verify valid user login                               |
| <b>Objective</b>       | To ensure only valid users can log in                 |
| <b>Precondition</b>    | User is registered with username and password         |
| <b>Test Steps</b>      | 1. Go to login page2. Enter credentials3. Click Login |
| <b>Input Data</b>      | username: admin, password: admin123                   |
| <b>Expected Result</b> | User is redirected to the dashboard                   |
| <b>Actual Result</b>   | As expected                                           |
| <b>Status</b>          | Pass                                                  |

```

System check identified 1 issue (0 silenced).
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Ran 1 test in 0.429s

OK
Destroying test database for alias 'default'...

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